

Assignment C

Common Notations

We use the following notation:

S_t	:	Stock price at time t
σ	:	Volatility of the Stock (assumed constant)
r	:	Interest rate
T	:	Time to option expiration (in years)
K	:	Strike price
W_t	:	Brownian motion process
$N(0, 1)$	:	Standard normal distribution
$(A - B)^+$	:	$\text{Max}(A-B, 0)$

Generating Random Inputs

- ▶ Where applicable, you may use the function shown below to generate input values randomly:

```
float random_data(float low, float hi)
{
    float r = (float)rand() / (float)RAND_MAX;
    return low + r * (hi - low);
}
```

- ▶ For example, you may use this function to populate the input matrices in Assignment 1, and initialize the input values in option pricing assignments.

Measuring Running Time using Chrono

- ▶ We use a timer to measure the execution times of a program.
- ▶ We have many choices to get such timing measurements.
Code snippet below shows how to use chrono in the C++ Standard Library.

```
#include <chrono>

using namespace std::chrono;

int main()
{
    high_resolution_clock::time_point t1 =
        high_resolution_clock::now();

    do_the_works();

    high_resolution_clock::time_point t2 =
        high_resolution_clock::now();

    std::cout << "Elapsed time: " <<
        duration_cast<milliseconds>(t2 - t1).count() << " ms";

}
```

- ▶ Defined in chrono header.

Assignment C(C/C++)

- ▶ Due: March 31 by 2 PM.
- ▶ Write a function to price European style Call options using a binomial tree.
- ▶ Jarrow-Rudd tree is explained below but you may use any other type (e.g. CRR) of your choice.
- ▶ Measure time taken to price 1 million options.
- ▶ You are not required to use techniques such as vectorization/multithreading for this assignment.
- ▶ The purpose of this assignment is to present the problem and encourage students to consider the performance aspects related to it.
- ▶ As long as anyone makes a genuine and honest attempt to solve this problem one will get full points for this assignment, even if the solution is not complete.

The Jarrow-Rudd Binomial Tree

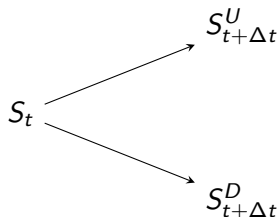
- ▶ We will use a binomial tree to simulate the paths of the underlying stochastic process.
- ▶ Jarrow-Rudd tree assumes the random value ($z=N(0, 1)$) takes $+1$ or -1 with equal (risk neutral) probability of $1/2$ at every node in the tree.
- ▶ If we know the stock price at time t , Equations 1 and 2 give us possible values for the stock prices at time $t + \Delta t$.

Stock Price Process

- This leads to stock price at time $t + \Delta t$:

$$z = 1 : S_{t+\Delta t}^U = S_t \exp((r - \sigma^2/2)\Delta t + \sigma\sqrt{\Delta t}) \quad (1)$$

$$z = -1 : S_{t+\Delta t}^D = S_t \exp((r - \sigma^2/2)\Delta t - \sigma\sqrt{\Delta t}) \quad (2)$$



- We refer to these two stock prices as $S_{t+\Delta t}^U$ and $S_{t+\Delta t}^D$ to mean the stock prices at *up* and *down* nodes respectively, at time $t + \Delta t$.

- ▶ We simulate how the stock prices would evolve on the tree using Equations 1 and 2:
 - ▶ divide time to expiration, T , into equal N parts such that $\Delta t = T/N$
 - ▶ discretize W_t such that

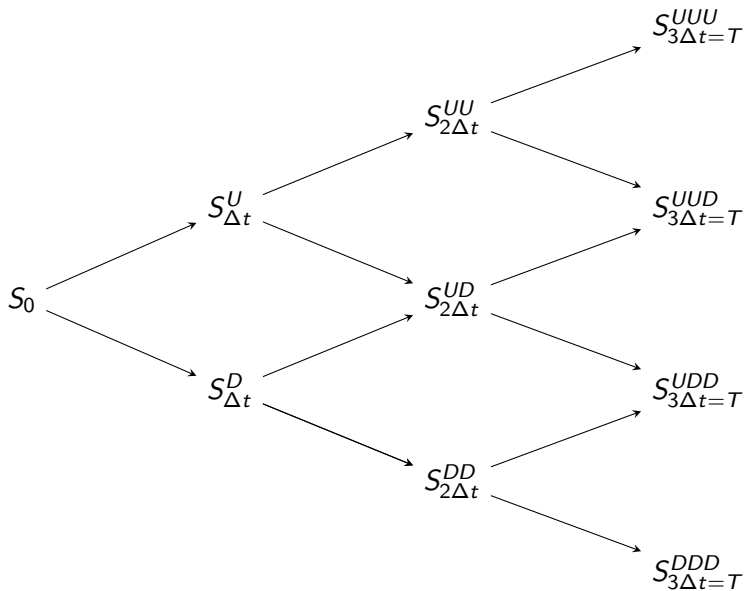
$$\Delta W_t = \sqrt{\Delta t} N(0, 1) \quad (3)$$

and assume the underlying Brownian motion can only move a fixed amount up or down at each time step.

(In the case of Jarrow-Rudd tree, the random variable $N(0, 1)$ given in Equation 3 takes $+1$ or -1 with an equal probability of $1/2$.)

- ▶ we start from the left of the tree, i.e. $t = 0$ and build the binary tree for the stock price process at every Δt increment

- A full binomial tree with 3 times steps ($N=3$) is shown next.



- ▶ Such a construction yields a binomial tree.
- ▶ This tree recombines by construction.
- ▶ Recombining means: if you start from a node at time t , an *up* move at time t followed by a *down* move at time $t + \Delta t$ and a *down* move at time t followed by an *up* move at time $t + \Delta t$ end up at the same node at time $t + 2\Delta t$

$$S_{\Delta t}^D = S_0 \exp((r - \sigma^2/2)\Delta t - \sigma\sqrt{\Delta t})$$

$$S_{\Delta t}^{DU} = S_{\Delta t}^D \exp((r - \sigma^2/2)\Delta t + \sigma\sqrt{\Delta t}) = S_0 \exp(2(r - \sigma^2/2)\Delta t) = S_{\Delta t}^{UD}$$

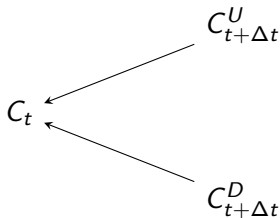
- ▶ Recombining trees do not grow very large when we increase the number of time steps.
- ▶ Note:
 1. The arrows show the direction of *path simulation*.
 2. Superscripts D, DD, U, UU etc. denote how many levels the stock price has gone up or down from the initial value.
U denotes an up movement; D denotes a down movement.

Pricing an European Option

We use the stock prices at each node on the tree to calculate the option price:

- ▶ The option payoff at time $t=T$, i.e. $C(S_T)$, is given by the options payoff function.
 - ▶ Call: $C(S_T) = \max(S_T - K, 0)$
 - ▶ Put: $C(S_T) = \max(K - S_T, 0)$
- ▶ The option payoff at time t , $C(S_t)$, is given by the discounted expectation of the option payoffs at time $t + \Delta t$:

$$C_t = [p_u C_{t+\Delta t}^U + p_d C_{t+\Delta t}^D] \exp^{-r\Delta t} \quad p_u = p_d = 0.5 \quad (4)$$



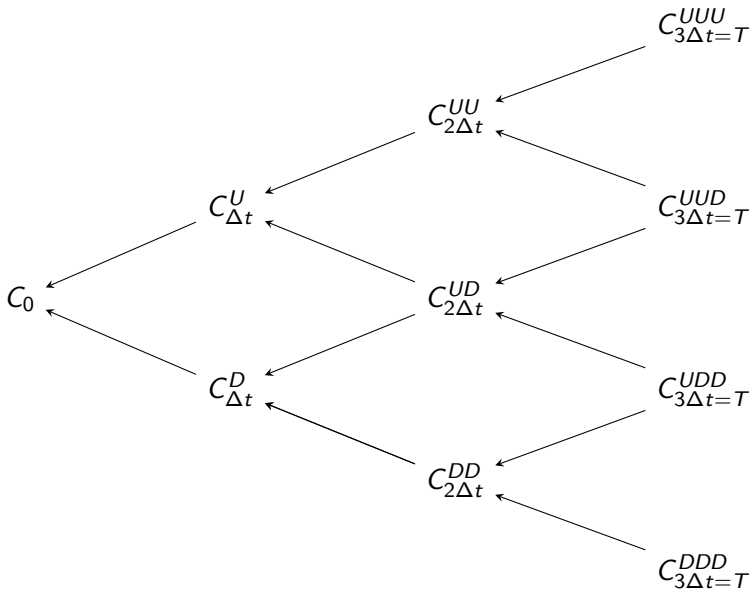
Pricing an European Option

- ▶ Risk neutral probabilities associated with each path (p_u and p_d) is 0.5 (by construction) for Jarrow-Rudd trees.
- ▶ Starting from the right ($t = T$) back-propagate the option prices to find the price of the option at time zero.
- ▶ The price of option at time t , i.e. C_t is given by the discounted expectation of prices at time $t + \Delta t$, given in Equation 4.

Notes:

1. The arrows show the direction of the calculation.
2. Superscripts D, DD, U, UU etc. correspond to the stock price levels.
3. 0.5 Risk neutral probability value in Equation 4 applies to Jarrow-Rudd trees only. Other types of trees may have different risk neutral probabilities.

- A full binomial tree with 3 time steps ($N=3$) is shown below:



- The option price at time zero ($t = 0$) is given by C_0 .